DO NOW — RECALL

Backtracking in computational solutions

Name: _____ Class: _____ Date: _____

1. When does a recursive call stop?

2. What is a search tree node?

3. Why may pruning reduce runtime?

4. Insert 12 into BST root10 with right child15 and left child5. Show the comparisons and position.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

When a base case is met.

2. Model answer

A partial choice/state considered in the search.

3. Model answer

It avoids exploring states proven unable to succeed.

4. Model answer

12>10 ⇒ go right to15; 12<15 ⇒ insert as left child of15.

By the end, you can…

- Make a decision
- Reject a dead end
- Backtrack

Vocabulary — know these terms

backtracking — Explore choices and undo a failed path.

pruning — Discard an impossible branch.

search tree — Branches of possible decisions.

base case — Condition ending a recursive branch.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain the method

Undo 4, skip 3, then include 4 → 5.

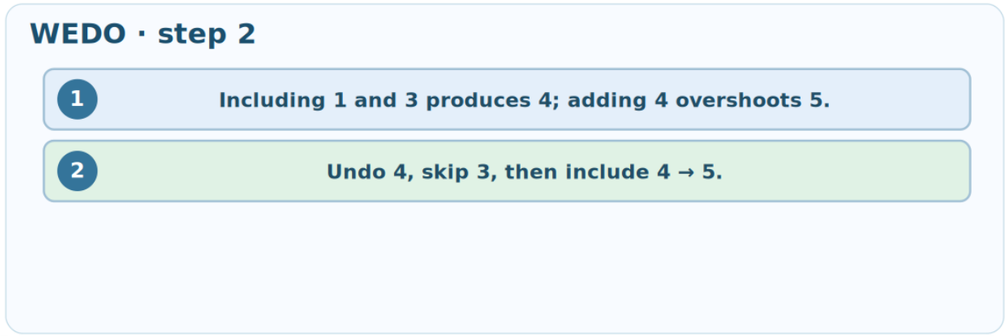
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find a subset totalling 5 from [1,3,4] using inclusion/exclusion.

Including 1 and 3 produces 4; adding 4 overshoots 5.



WEDO · step 2; Including 1 and 3 produces 4; adding 4 overshoots 5.; Undo 4, skip 3, then include 4 → 5.

NEW LEARNING

Read, annotate and explain

Make a decision

At each position choose the current item or skip it. Track index and running total.

Reject a dead end

When the sum exceeds the target (non-negative inputs) or the list ends, undo the choice.

Backtrack

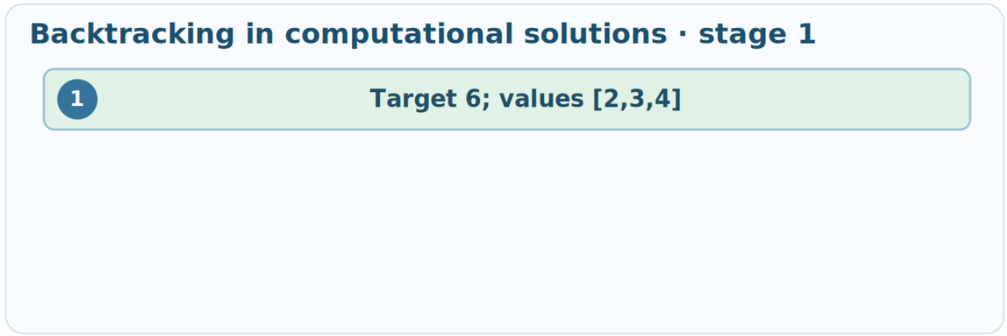
Try the other branch; success means a valid selection has been found. Exponential worst-case growth remains possible.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

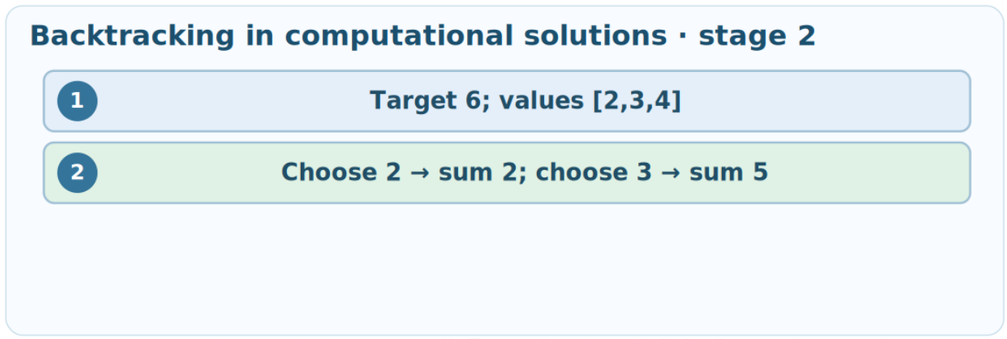
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Backtracking in computational solutions · stage 1; Target 6; values [2,3,4]

New Learning



Backtracking in computational solutions · stage 2; Target 6; values [2,3,4]; Choose 2 → sum 2; choose 3 → sum 5

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Backtracking can require trying an earlier alternative after a dead end.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

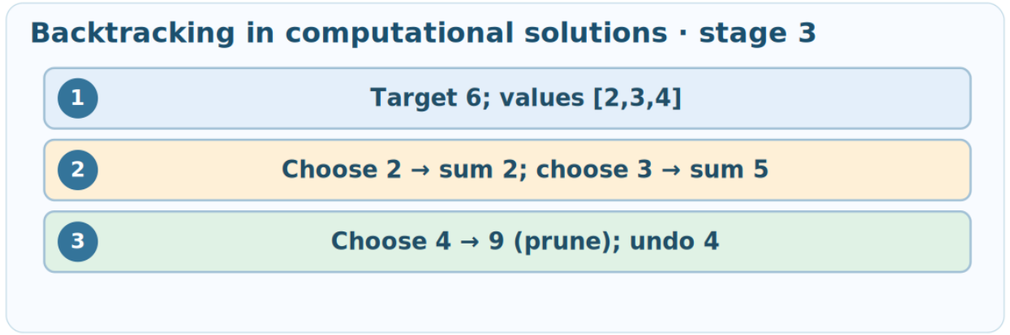
Question 3

Explain why backtracking is useful for 4 queens but not guaranteed polynomial time. Give one safe-placement rule. [6]

New Learning · original lesson diagrams

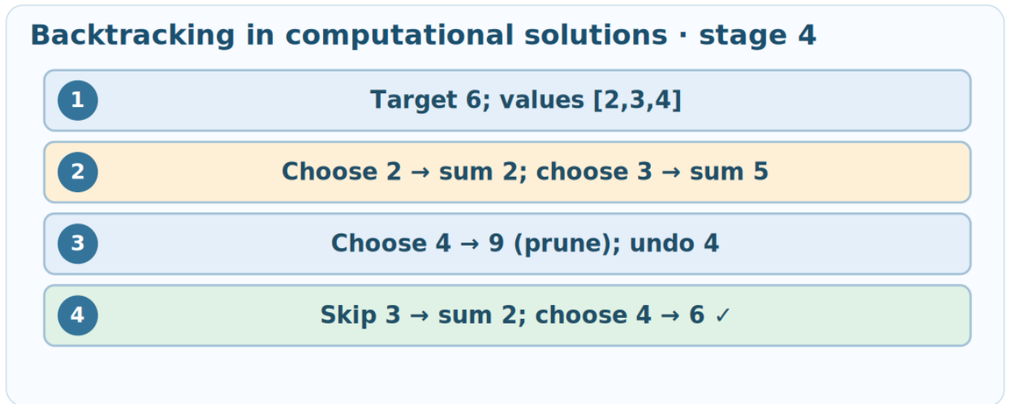
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Backtracking in computational solutions · stage 3; Target 6; values [2,3,4]; Choose 2 → sum 2; choose 3 → sum 5; Choose 4 → 9 (prune); undo 4

New Learning



Backtracking in computational solutions · stage 4; Target 6; values [2,3,4]; Choose 2 → sum 2; choose 3 → sum 5; Choose 4 → 9 (prune); undo 4; Skip 3 → sum 2; choose 4 → 6 ✓

I DO — WORKED EXAMPLE

Follow the worked method

Find a subset totalling 6 from [2,3,4]. Trace the first failed path and successful branch.

Predict first

The first path over-shoots: 2+3+4=9, so reject that branch.

Explain the method

Undo 4, skip 3, then include 4 → 6.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain choice, dead end and backtrack using [2,3,4] target6. [6]

Question 2

Write a recursive include/exclude Boolean function for subset sum with non-negative integers, including base cases. [8]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Including 1 and 3 produces 4; adding 4 overshoots 5.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

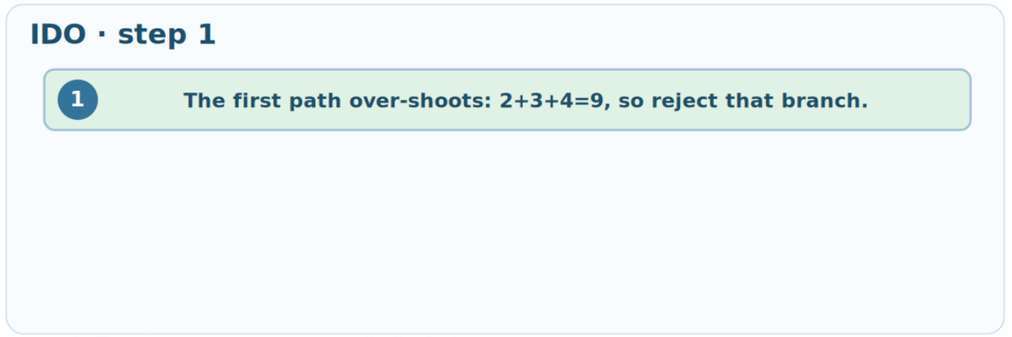
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find a subset totalling 6 from [2,3,4]. Trace the first failed path and successful branch.

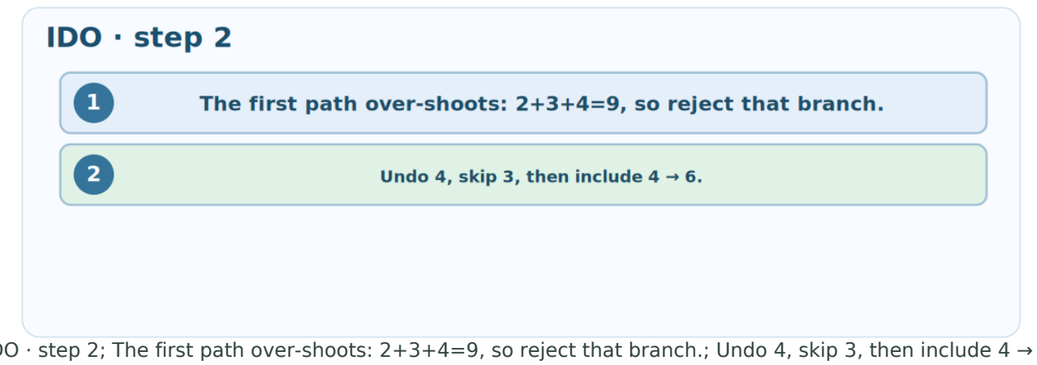
The first path over-shoots: 2+3+4=9, so reject that branch.



IDO · step 1; The first path over-shoots: 2+3+4=9, so reject that branch.

Find a subset totalling 6 from [2,3,4]. Trace the first failed path and successful branch.

The first path over-shoots: 2+3+4=9, so reject that branch.



IDO · step 2; The first path over-shoots: 2+3+4=9, so reject that branch.; Undo 4, skip 3, then include 4 → 6.

WE DO — GUIDED PRACTICE

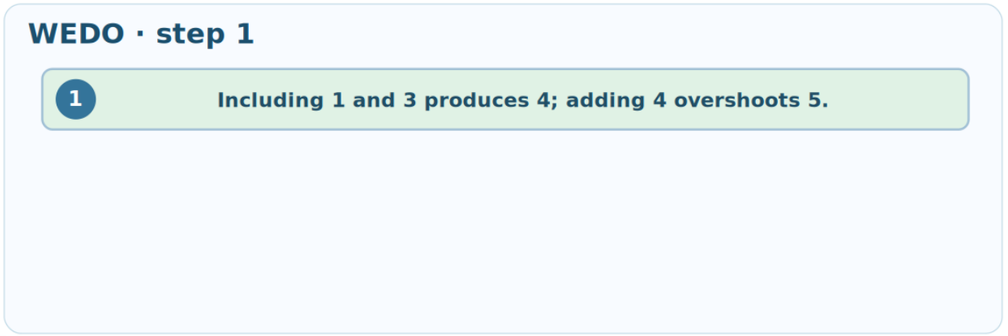
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find a subset totalling 5 from [1,3,4] using inclusion/exclusion.

Including 1 and 3 produces 4; adding 4 overshoots 5.



WEDO · step 1; Including 1 and 3 produces 4; adding 4 overshoots 5.